# PRITOM MAJUMDER

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## RESEARCH INTEREST

- **Functional & Piezoelectric Ceramics:** Lead-free ferroelectrics, solid-state synthesis, dielectric and piezoelectric response, magnetic ceramics and low-temperature sintering routes.
- **Oxide Semiconductor Thin Films:** Doping-driven control of structural, optical, magnetic and electronic behaviour in transparent conducting oxides.
- **Photonic & Optoelectronic Materials:** Advance ceramics, lasers, LEDs, photovoltaics, photonics and magnetic data storage and optical fibers.
- **Metallurgy:** Corrosion and surface engineering, ferrous metallurgy, materials characterization and mechanical behaviour of materials.

## EDUCATION

**B.Sc. in Materials and Metallurgical Engineering** **2021 to June 2026**

Bangladesh University of Engineering and Technology (BUET), Dhaka

**CGPA:** 3.64 ($15^{th}$ of 59) / 4.00

**Relevant coursework:** Advanced Ceramics (A+); Nanostructured Materials & Thin Films (A+); Electrical, Optical and Magnetic Properties of Materials (A+); Characterisation of Materials Laboratory (A+); Photonic and Magnetic Materials (A); Characterisation of Materials (A−); Crystallography and Structure of Materials (A−); Phase Diagrams & Transformations (A−).

## ON GOING WORK

1. **Pritom Majumder**, Md. Atik Faisal, Mehrab Rahman, Anabil Dey, and Md. Azizar Rahman. “Structural, Optical, Magnetic, and Electronic Properties of Hydrothermally Grown Cu-Doped $SnO_2$ Thin Films: A Combined Experimental and First-Principles (DFT+U) Study.” Manuscript in preparation.
2. **Pritom Majumder**, Anabil Dey, and Ahmed Sharif. “Advancements in Cold Sintering of Piezoceramics: Flux Mediation, Microstructural Evolution, and Enhanced Dielectric Properties - A Review.” Manuscript in preparation.

## RESEARCH EXPERIENCE

**Undergraduate Thesis Researcher** · Ahmed Sharif’s Lab · Department of MME, BUET **2025 to 2026**

“Investigation of Dielectric and Piezoelectric Properties of KNN with BCHfT Composite Ceramics” ; Supervisor: Prof. Ahmed Sharif ; THESIS BOOK ; THESIS PPTX

- Synthesized lead-free $K_{0.5}Na_{0.5}NbO_3$ (KNN) ceramics modified with $Ba_{0.85}Ca_{0.15}Hf_{0.1}Ti_{0.9}O_3$ (BCHfT) by the conventional solid-state route.
- Characterized phase evolution and microstructure by XRD and FE-SEM; measured frequency- and temperature-dependent dielectric permittivity and loss, impedance spectroscopy, and the piezoelectric charge coefficient $d_{33}$.
- Identified KNN-1wt% BCHfT as the optimum composition, giving the best balance of dielectric response.

**Undergraduate Thesis Researcher** · Solid State Physics Laboratory · Department of Physics, BUET **2024 to 2026**

- Synthesis of SnO2 and Cu:SnO2 research, Characterisation: PL, UV-Vis, I-V, C-V, Mott-Schottky.

**Undergraduate Thesis Researcher** · Pilot plant and Process Development Center · BCSIR **2025 to 2026**

- Carried out the complete characterization of the KNN-BCHfT thesis series on BCSIR instruments: XRD, SEM, frequency- and temperature-dependent dielectric test

## SELECTED PROJECTS

**Millsorpt - Capstone Design Project** · Department of MME, BUET **2025**

- Converted ferrous mill scale, a low-value steel-industry waste, into a chemically treated adsorbent(hematite rich iron-oxide) for heavy metal wastewater treatment

- Designed a low-cost bench scale filtration column achieving a high removal of Cu, Cr and Ni from mixed-metal solutions; iron-oxide phases confirmed by XRD, metal removal quantified by AAS and UV-Vis.
- Presented the work to specialists at Abul Khair Steel, BCSIR, the Bangladesh Atomic Energy Commission etc., receiving positive technical feedback

**3D PotterBot - CAD Modelling** · SolidWorks **2023**

- Modelled a paste-extrusion ceramic 3D printer as a fully-mated 118 parts assembly; produced photorealistic renders, an exploded-view motion study, and a static Finite Element Analysis of the vertical support rod

## INDUSTRIAL AND LABORATORY TRAINING

**Bangladesh Council of Scientific and Industrial Research (BCSIR)** · Dhaka **2026**

- Performed the characterization for my undergraduate thesis on BCSIR instruments: XRD, SEM, and temperature- and frequency-dependent dielectric measurement

**Abul Khair Steel Melting Limited (AKSML)** · Chittagong **March 2025**

- Studied electric arc furnace, ladle refining furnace and continuous casting operations, and rolling-mill practice using Quenching & Tempering Bar (QTB) technology for structural steel.
- Observed quality-control workflows spanning ICP-OES, XRF, handheld XRF and wet chemical analysis, together with process automation and real-time monitoring

**Bangladesh Industrial Technical Assistance Centre (BITAC)** · Dhaka **March 2025**

- Hands-on training in precision machining (lathe, milling, CNC), TIG/arc/spot welding, heat treatment, foundry operations, CAD-CAM and industrial automation

## TECHNICAL SKILLS

- **Synthesis, Processing and Data Analysis:** Solid-state ceramic synthesis; hydrothermal synthesis; sintering; uniaxial pressing; XRD; FE-SEM and SEM; TEM including HRTEM, SAED; Impedance spectroscopy; frequency- and temperature-dependent dielectric measurement; $d_{33}$ meter; cyclic voltammetry; I-V and Mott-Schottky analysis; UV-Vis spectroscopy; photoluminescence (PL); FTIR
- **Mechanical & Chemical Testing:** Tensile, Charpy impact and three-point flexural testing; Vickers, Rockwell and Brinell hardness; optical emission spectroscopy (OES); gravimetric and wet chemical analysis
- **Software:** OriginPro; MATLAB; SolidWorks; ImageJ; X'Pert HighScore; C/C++; MS Office; Adobe Photoshop and Illustrator

## AWARDS AND SCHOLARSHIPS

- **University Merit Scholarship**, Bangladesh University of Engineering and Technology **(July 2022)**
- **University Stipend Scholarship**, Bangladesh University of Engineering and Technology (July 2021 and January 2022)
- **Letter of Appreciation**, Abul Khair Steel Melting Limited - for professionalism and work ethic during industrial training **2025**
- **Certificate of Perfect Attendance**, Notre Dame College, Dhaka **(2019 to 2020)**

## LANGUAGE PROFICIENCY

**English:** IELTS Academic - Overall Band 7.0 (Listening 7.5, Reading 7.5, Writing 6.0, Speaking 6.5) **2026**

## LEADERSHIP AND EXTRACURRICULAR ACTIVITIES

- **Director of Administration**, Executive Committee, BUET Debating Club (BUETDC), 2025-26 :- previously Information Secretary. Organizing panel member for BUETDC Nationals 2024; finalist at the BUET Freshers' Debating Tournament; competed at the Inter-Faculty Tournament 2024 and the 6th GSCDC Nationals 2024.
- **Organizer**, Winter Fitness Competition 2025, BUET Gymnasium :- planned and ran the event in collaboration with the resident trainer; Organized Treasure-Hunt competition 2024, as an organizing panel member of "Reception'21"
- **Class Representative**, Department of MME, BUET, 2021-22 :- elected by classmates for both semesters of the first year
- **Active Member & Blood Donor**, Badhon - a non-profit voluntary blood donation organization, 2022-present